Homework for week 8

[The notation D is used for the derivative operator, I is the identity, explanation in section 2.3. Some of these we already solved in class.]

1. 2.7 Problems 1, 3, 4, 5, 7, 11, 12, 14, 18
2. 3.2 Problems 1, 2, 6, 10
3. 3.3 Problem 1
4. *Bending Beam (Kreyszig 3.3)*

   The shape of a horizontal uniform elastic beam of length L bending under its own weight is described by the ODE:

   $$\frac{d^4y}{dx^4} = k$$

   Solve the initial value problem in various cases and sketch the solution:

   a) Both ends clamped:

   $$y(0) = y'(0) = 0, \;\; y(L) = y'(L) = 0$$

   b) One end clamped, one free:

   $$y(0) = y'(0) = 0, \;\; y''(L) = y'''(L) = 0$$

   c) One end clamped, one hinged:

   $$y(0) = y'(0) = 0, \;\; y(L) = y''(L) = 0$$

   d) Bonus: $2L$ long beam, one end at x=-L clamped, hinged support at x=0 and free end at x=L. [Write two general solutions with independent parameters for the two parts of the beam, $y_1(x)$ for [-L, 0] and $y_2(x)$ for [0, L]. Use the usual boundary conditions at L and -L. At the hinged support at 0 use $y_1(0) = y_2(0) = 0$, $y_1'(0) = y_2'(0)$, $y_1''(0) = y_2''(0)$. Why are these the correct boundary conditions?]